

Tae Hong (Richard) Kim

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Technical Skills

- **Languages:** Java, Python, TypeScript/JavaScript, C#, SQL, C, C++, PowerShell, YAML, Bash, HTML/CSS
- **Frameworks & Data:** Spring Boot, React, Next.js, Angular, ASP.NET Core/.NET, PostgreSQL, Oracle DB
- **Cloud & DevOps:** Azure, AWS, AKS, APIM, Docker, Kubernetes, Helm, Terraform, GitHub Actions, Postman

Experience

Toronto Dominion Bank (TD)

Sept 2026 – Present

Software Engineer Intern

Toronto, ON, Canada

- Engineering **C#/ASP.NET** Core validation service and **Angular dashboard** that processes system change requests, resolve application-specific validation rules, and *orchestrate* **GitHub Copilot SDK** inference into structured JSON responses, *reducing manual document review by 60%*.
- Building **CI/CD** pipelines for **C#/ASP.NET** and **Angular** applications using **GitHub Actions**, automating builds, **SonarQube** quality gates, and **Nexus** artifact publishing prior to orchestrating *environment configuration* and *server deployments* through **Digital.ai** Release/Deploy, targeting an **80%** reduction in manual deployment effort.
- Developing an *automated validation workflow* across TD Wealth data pipelines to detect inactive user accounts before downstream API execution, reducing manual checks and failed processing using **PowerShell**, **AutoSys**, and **SQL Server**.

FundServ Inc

Jan 2026 – April 2026

Software Developer Intern

Toronto, ON, Canada

- Architected and deployed *secure microservices* to **AKS** with **Terraform**, **Helm**, and **Azure APIM**, implementing **MSAL**, **PKCE**, and **JWT** security policies for *CIRO (Canadian Investment Regulatory Organization)* compliant *B2B communication*
- Built **GitOps CI/CD pipeline** with **GitHub Actions**, **Google Jib**, **ACR**, and **ArgoCD**, automating *containerized service deployments*, reducing release effort by **70%**
- Integrated **Veracode**, **Prisma**, **Vault**, and **JaCoCo** into **YAML** build workflows, enforcing *security scanning*, *secret management*, and **80% unit-test coverage** before production deployment

Software Developer Intern

Jan 2025 – April 2025

- Engineered *full-stack onboarding microservice* with **Java**, **Spring Boot**, **TypeScript**, **Angular**, and **Oracle DB**, automating manual support workflows for financial institution onboarding
- Developed *CSV ingestion and verification service* with **Spring JPA** and **SQL**, processing financial data from *Unix file servers* and *reducing onboarding errors by 40%*
- Led technical design reviews by producing *HLD/LLD documentation* and *end-to-end workflow diagrams*, securing CIO approval for production deployment

Ford Pro

Sep 2023 – Dec 2023, May 2024 – Aug 2024

Exploratory Test Engineering Intern

Oakville, ON, Canada

- Optimized **CI/CD pipelines** utilizing **Tekton**, **Docker**, and **Azure DevOps** by implementing job parallelization and **Linux** shell caching strategies, maximizing build efficiency and deployment reliability for distributed production services
- Engineered automated validation frameworks using **Playwright**, **SoapUI**, and **Postman** to verify **SQL database** operations and *API integrity*, accelerating *feedback loops* by **50%** while monitoring *system health* via **Datadog**

Projects

AI Website Generator - Next.js, TypeScript, OpenRouter, Drizzle, PostgreSQL

 [Th32kim/AlwebGen](#)

- Architected a *Generative UI engine* utilizing **Next.js** and **OpenRouter APIs**, implementing **Server-Sent Events (SSE)** and system prompting to enforce structured code generation for real-time streaming of interactive components via *iframe*
- Developed a *high-performance persistence layer* with **Drizzle ORM** and **Neon PostgreSQL** to version control AI generated configurations, ensuring type-safe, sub-second UI rendering through **TypeScript** and **Tailwind CSS**

AI-Agent Search Engine - Python, LangGraph, LangChain, OpenAI API, Pydantic

 [Th32kim/AI-Agent](#)

- Engineered a **LangGraph**-orchestrated data pipeline using Bright Data async dataset triggers and snapshot polling to move long-running web and Reddit ingestion outside the interactive LLM request path.
- Implemented **LangChain**, **OpenAI** multi pass reasoning over Google, Bing, and Reddit artifacts while enforcing **Pydantic** structured URL lists for targeted Reddit follow ups and synthesized **LLM** and *web-scraped* outputs into single result

Education

University of Waterloo

Sep 2022 – May 2027

Bachelor of Applied Sciences - Computer Engineering Honours Co-op

Waterloo, Ontario, Canada

Courses: **Computational Intelligence (ML)**, **Database Systems (SQL)**, **Systems Programming and Concurrency (C)**